Arda YILDIZ

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Education

Çankaya University B.Sc. in Computer Engineering (60% Scholarship) CGPA: 3.23 — Sep 2021 – June 2026

Experience

NanoMagnetics Instruments — Feb 2026 – Present
Candidate Engineer
- Developing embedded and software solutions; contributing to hardware-software integration and system communication.

Çankaya University — Mar 2024 – June 2026
IT Department – Part-Time Student
- Working as a part-time developer in the IT Department, contributing to university-level system development projects.
- Collaborating with a teammate to design and implement real-world embedded and software systems.
- Designed and developed the following systems in collaboration with a teammate:
  - **Smart School Bus Tracking System:** Developed a smart bus access and tracking system for Çankaya University using an ESP32 microcontroller and SIM7600 4G communication module, integrated with university servers via a REST API; assigned by the Board of Trustees of the university.
  - **RFID-Based University Access Control System:** Developed a production-level RFID-based access control system actively used at Çankaya University; assigned by the Board of Trustees of the university.

Digitest Savunma ve Havacılık A.Ş. — Jun 2025 – Sep 2025
Long-Term Embedded Systems Intern
- Designed FPGA systems in Vivado with custom IP cores and AXI-based communication; developed embedded software in bare-metal and FreeRTOS environments.
- Implemented UART and TCP/IP communication pipelines for real-time data exchange; trained YOLOv5 models and built a PySide6 GUI for real-time object detection.

NanoMagnetics Instruments — Jul 2024 – Sep 2024
Long-Term Software Intern
- Developed a Flutter-based mobile application to control scientific devices via BLE and API; designed scalable architecture using Provider state management and modular API structure.
- Integrated Firebase Authentication and Firestore for dynamic content and user management; optimized performance by identifying memory leaks and improving resource usage.

Projects

Helma AI – AI-Powered Elderly Care Assistant (helma-ai.com) — Oct 2025 – June 2026
Developing a multimodal AI-powered mobile application for elderly care that implements emotion recognition using video, audio, and text inputs. The system is built with Flutter, FastAPI, and MongoDB, and includes real-time monitoring, reminders, and caregiver-family interaction workflows.

Turkish Legal Question Answering System (github/29ardayildiz/turkish-legal-rag-pipeline) — Feb 2026 – June 2026
Developing an end-to-end RAG system for Turkish legal question answering, including embedding tuning, hybrid retrieval, reranking, and LLM fine-tuning. Designed to produce grounded, citation-consistent answers while minimizing hallucination, evaluated using retrieval and QA metrics such as Recall@K, MRR, F1, and faithfulness.

TinyVoice – Keyword Spotting System (github/ackgz0/tinyVoiceKWS) — Nov 2025 – Jan 2026
Developed an edge AI-based keyword spotting system optimized for microcontrollers by training and comparing multiple deep learning models on the Google Speech Commands dataset, and preparing a DS-CNN model for deployment on ESP32-class devices.

Push Or Perish (github/29ardayildiz/Push_Or_Perish) — Oct 2024 – Jan 2025
Developed a reflex-based game on the MSP430G2553 using a 7-segment display, two buttons, and two LEDs. Implemented countdown control, input-based winner logic, LED indication, and automatic game restart in an embedded systems environment.

VehiCrypto-Blockchain-Based Vehicle Maintenance Tracking (github.com/dogukanpoyraz/VehiCrypto) — Oct 2024 – Dec 2024
Designed a decentralized vehicle maintenance tracking system using blockchain and IPFS technologies to improve transparency, security, and data integrity.

Skills

Programming: C, C++, Java, Assembly, Python, C#, Dart, JavaScript, SQL
AI / Machine Learning: TensorFlow, PyTorch, OpenCV, RAG, LLM Fine-Tuning
Embedded Systems: ESP32, MSP430, FPGA (Vivado), FreeRTOS
Mobile & Backend: Flutter, FastAPI, MongoDB, REST APIs
Tools: Git, Linux/Bash, Arduino, Qt Creator, Unity, Unreal Engine, Vitis
Communication Protocols: UART, SPI, I2C, I2S, TCP/IP, UDP, HTTP/REST, Wi-Fi, Bluetooth

Additional

Languages: Turkish (Native), English (B2)
Certificates: METU Robotics Days Participation Certificate; AI Tomorrow Summit 2023; HackerRank Python (Basic)
Activities: Google Developer Student Clubs (Member, 2021 – 2024), IEEE Çankaya (Algorithm Team Member, 2021 – 2023)
Personal Traits: Team player, analytical thinker, creative, fast learner